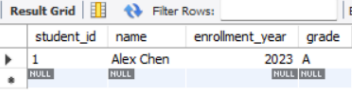


## Task 1: Basic Insertion

- Create a table named students with columns: student\_id (INT, PRIMARY KEY, AUTO\_INCREMENT), name (VARCHAR(100)), enrollment\_year (YEAR), grade (CHAR(1)).
- Insert a record for a student named "Alex Chen" enrolled in 2023 with a grade of 'A'.

```
5 • create table students
6   ( student_id INT PRIMARY KEY auto_increment,
7     name varchar(100),
8     enrollment_year year,
9     grade char(1))
10
11 • insert into students(name,enrollment_year,grade) values ("Alex Chen",2023,"A");
```

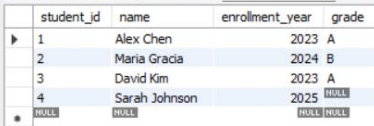


| student_id | name      | enrollment_year | grade |
|------------|-----------|-----------------|-------|
| 1          | Alex Chen | 2023            | A     |

## Task 2: Multiple Row Insertion

- Using the students table, insert three new students in one INSERT statement:
  - "Maria Garcia", 2024, 'B'
  - "David Kim", 2023, 'A'
  - "Sarah Johnson", 2025, NULL

```
14 • insert into students (name,enrollment_year,grade) values('Maria Garcia',2024,'B'),
15   ('David Kim',2023,'A'),
16   ('Sarah Johnson', 2025, NULL);
17 • select * from students;
```



| student_id | name          | enrollment_year | grade |
|------------|---------------|-----------------|-------|
| 1          | Alex Chen     | 2023            | A     |
| 2          | Maria Garcia  | 2024            | B     |
| 3          | David Kim     | 2023            | A     |
| 4          | Sarah Johnson | 2025            | NULL  |

## Task 3: Insert with Specific Columns (Handling NULL)

Insert a new student named "Tommy Lee" into the students table. You only know his student\_id should be 100 and his enrollment\_year is 2025. The grade is unknown. Write the statement without causing an error.

```

19 • insert into students(student_id,name,enrollment_year) values (100,'Tommy Lee',2025);
20 • select *from students;
21
22

```

| Result Grid                             |               |                 |           |  |
|-----------------------------------------|---------------|-----------------|-----------|--|
| Filter Rows:                            |               |                 |           |  |
| Edit: Export/Import: Wrap Cell Content: |               |                 |           |  |
| student_id                              | name          | enrollment_year | grade     |  |
| 1                                       | Alex Chen     | 2023            | A         |  |
| 2                                       | Maria Gracia  | 2024            | B         |  |
| 3                                       | David Kim     | 2023            | A         |  |
| 4                                       | Sarah Johnson | 2025            | NULL      |  |
| 100                                     | Tommy Lee     | 2025            | NULL      |  |
| NULL                                    | NULL          |                 | NULL NULL |  |

## Task 4: Basic Update (Single Condition)

Sarah Johnson (inserted in Task 2) has received her grade. Update her record to set her grade to 'B+'.

```

24 • create index id_name on students(name);
25 • update students set grade='B+' where name= 'Sarah Johnson';
26 • select *from students;
27
28
29
30

```

| Result Grid                             |               |                 |           |  |
|-----------------------------------------|---------------|-----------------|-----------|--|
| Filter Rows:                            |               |                 |           |  |
| Edit: Export/Import: Wrap Cell Content: |               |                 |           |  |
| student_id                              | name          | enrollment_year | grade     |  |
| 1                                       | Alex Chen     | 2023            | A         |  |
| 2                                       | Maria Gracia  | 2024            | B         |  |
| 3                                       | David Kim     | 2023            | A         |  |
| 4                                       | Sarah Johnson | 2025            | B         |  |
| 100                                     | Tommy Lee     | 2025            | NULL      |  |
| NULL                                    | NULL          |                 | NULL NULL |  |

## Task 5: Conditional Update (Multiple Conditions)

In the students table, update the grade to 'A+' for all students who are enrolled in 2023 and currently have a grade of 'A'.

```

29 • select *from students;
30
31 • ALTER TABLE students
32   MODIFY grade CHAR(2);
33
34 • update students set grade= 'A+' where enrollment_year=2023 and grade= 'A';
35 • select * from students;

```

| Result Grid                             |               |                 |           |  |
|-----------------------------------------|---------------|-----------------|-----------|--|
| Filter Rows:                            |               |                 |           |  |
| Edit: Export/Import: Wrap Cell Content: |               |                 |           |  |
| student_id                              | name          | enrollment_year | grade     |  |
| 1                                       | Alex Chen     | 2023            | A+        |  |
| 2                                       | Maria Gracia  | 2024            | B         |  |
| 3                                       | David Kim     | 2023            | A+        |  |
| 4                                       | Sarah Johnson | 2025            | B+        |  |
| 100                                     | Tommy Lee     | 2025            | NULL      |  |
| NULL                                    | NULL          |                 | NULL NULL |  |

### Task 6: Update All Rows (Cautionary Task)

Write a statement that would set the enrollment\_year to 2024 for **every record** in the students table. (The tester should recognize the danger of omitting the WHERE clause).

```
37 • update students set enrollment_year=2024;
38 • select * from students;
```

20

Result Grid

| student_id | name          | enrollment_year | grade |
|------------|---------------|-----------------|-------|
| 1          | Alex Chen     | 2024            | A+    |
| 2          | Maria Gracia  | 2024            | B     |
| 3          | David Kim     | 2024            | A+    |
| 4          | Sarah Johnson | 2024            | B+    |
| 100        | Tommy Lee     | 2024            | NULL  |
| NULL       | NULL          | NULL            | NULL  |

### Task 7: Basic Deletion (Single Condition)

Delete the record for "David Kim" from the students table.

```
40 • delete from students where name='David Kim';
41 • select * from students;
```

42

Result Grid

| student_id | name          | enrollment_year | grade |
|------------|---------------|-----------------|-------|
| 1          | Alex Chen     | 2024            | A+    |
| 2          | Maria Gracia  | 2024            | B     |
| 4          | Sarah Johnson | 2024            | B+    |
| 100        | Tommy Lee     | 2024            | NULL  |
| NULL       | NULL          | NULL            | NULL  |

### Task 8: Conditional Deletion (IN operator)

Some students have dropped out. Delete all students whose enrollment\_year is either **2024** or **2025**.

```
43 • delete from students where enrollment_year in (2024,2025);
44 • select * from students;
```

45

Result Grid

| student_id | name | enrollment_year | grade |
|------------|------|-----------------|-------|
| NULL       | NULL | NULL            | NULL  |

### Task 9: Safe Deletion vs. Truncation

Write a statement to remove all rows from the students table but leave its structure intact for future use. Is this DELETE or TRUNCATE?

```

45
46 • truncate table students;
47 • select * from students;

```

AR

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

| student_id | name | enrollment_year | grade |
|------------|------|-----------------|-------|
| NULL       | NULL | NULL            | NULL  |

## Task 10: Integrated Task – Real-World Scenario

Use a new book\_inventory table:

1. **Insert:** Add three books to the inventory.

```

58
59 • insert into book_inventory values(1,'Outdated Edition','Alexa Samuvel',15,200.50),
60 (2,'The Great Novel','Shwetha George',4,500.00),
61 (3,'His First Flight','author A',10,150.00);
62 • select * from book_inventory;

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

| book_id | title            | author         | quantity | price  |
|---------|------------------|----------------|----------|--------|
| 1       | Outdated Edition | Alexa Samuvel  | 15       | 200.50 |
| 2       | The Great Novel  | Shwetha George | 4        | 500.00 |
| 3       | His First Flight | author A       | 10       | 150.00 |
| NULL    | NULL             | NULL           | NULL     | NULL   |

2. **Update:** The book titled "The Great Novel" just arrived in a new shipment. Increase its quantity by 10.

```

75
76 • UPDATE book_inventory
77 SET quantity = quantity + 10
78 WHERE title = 'The Great Novel';
79 • select * from book_inventory;

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

| book_id | title            | author         | quantity | price  |
|---------|------------------|----------------|----------|--------|
| 1       | Outdated Edition | Alexa Samuvel  | 15       | 200.50 |
| 2       | The Great Novel  | Shwetha George | 14       | 500.00 |
| 3       | His First Flight | author A       | 10       | 150.00 |
| NULL    | NULL             | NULL           | NULL     | NULL   |

3. **-Update:** Due to a pricing error, reduce the price of all books written by "Author A" by 10%.

```
81 • UPDATE book_inventory
82   SET price = price -(price * 0.10)
83   WHERE author = 'author A';
84 • select * from book_inventory;
```

| Result Grid                             |                  |                |          |        |  |
|-----------------------------------------|------------------|----------------|----------|--------|--|
| Filter Rows:                            |                  |                |          |        |  |
| Edit: Export/Import: Wrap Cell Content: |                  |                |          |        |  |
| book_id                                 | title            | author         | quantity | price  |  |
| 1                                       | Outdated Edition | Alexa Samuel   | 15       | 200.50 |  |
| 2                                       | The Great Novel  | Shwetha George | 14       | 500.00 |  |
| 3                                       | His First Flight | author A       | 10       | 135.00 |  |
| •                                       | NULL             | NULL           | NULL     | NULL   |  |

4. **Delete:** A book titled "Outdated Edition" has been discontinued and must be removed from the inventory table.

```
85
86 • delete from book_inventory where title = 'Outdated Edition';
87 • select * from book_inventory;
```

| Result Grid                             |                  |                |          |        |  |
|-----------------------------------------|------------------|----------------|----------|--------|--|
| Filter Rows:                            |                  |                |          |        |  |
| Edit: Export/Import: Wrap Cell Content: |                  |                |          |        |  |
| book_id                                 | title            | author         | quantity | price  |  |
| 2                                       | The Great Novel  | Shwetha George | 14       | 500.00 |  |
| 3                                       | His First Flight | author A       | 10       | 135.00 |  |
| •                                       | NULL             | NULL           | NULL     | NULL   |  |